Quiz 4 - DSC 10, Fall 2023

Full Name:

PID:

Discussion: ◯ A (3PM) ◯ B (4PM) ◯ C (5PM)

Instructions:

- This quiz consists of 3 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles; if we cannot tell which option(s) you selected, you may lose points. For multiple choice questions, select **only one answer**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- No aids are allowed (no notes, no calculators, and no computers).

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

Video streaming services, such as Netflix and Hulu, are more popular than ever. Let's investigate trends in the video streaming habits of UCSD students.

Question 1

You survey 100 DSC majors and 140 CSE majors to ask them which video streaming service they use most. The resulting distributions are given in the table below. Note that each column sums to 1.

Service	**DSC Majors**	**CSE Majors**
Netflix	0.4	0.35
Hulu	0.25	0.2
Disney+	0.1	0.1
Amazon Prime Video	0.15	0.3
Other	0.1	0.05

For example, 20% of CSE Majors said that Hulu is their most used video streaming service. Note that if a student doesn't use video streaming services, their response is counted as Other.

a) What is the total variation distance (TVD) between the distribution for DSC majors and the distribution for CSE majors? Give your answer as an exact decimal.

b) Suppose we only break down video streaming services into four categories: Netflix, Hulu, Disney+, and Other (which now includes Amazon Prime Video). Now we recalculate the TVD between the two distributions. How does the TVD now compare to your answer to part (a)?

◯ less than (a) ◯ equal to (a) ◯ greater than (a)

Question 2

You want to estimate the proportion of DSC majors who have a Netflix subscription. To do so, you will survey a random sample of DSC majors and ask them whether they have a Netflix subscription. You will then create a 95% confidence interval for the proportion of "yes" answers in the population, based on the responses in your sample. You decide that your confidence interval should have a width of at most 0.10.

a) In order for your confidence interval to have a width of at most 0.10, the standard deviation of the distribution of the sample proportion must be at most T. What is T? Give your answer as an exact decimal.

b) Using the fact that the standard deviation of any dataset of 0s and 1s is no more than 0.5, calculate the minimum number of people you would need to survey so that the width of your confidence interval is at most 0.10. Give your answer as an integer.

Question 3

Arya was curious how many UCSD students used Hulu over Thanksgiving break. He surveys 250 students and finds that 130 of them did use Hulu over break and 120 did not.

Using this data, Arya decides to test following hypotheses:

- **Null Hypothesis**: Over Thanksgiving break, an equal number of UCSD students did use Hulu and did not use Hulu.
- **Alternative Hypothesis**: Over Thanksgiving break, **more** UCSD students did use Hulu than did not use Hulu.

a) Which of the following could be used as a test statistic for the hypothesis test?

◯ The proportion of students who did use Hulu minus the proportion of students who did not use Hulu.

◯ The absolute value of the proportion of students who did use Hulu minus the proportion of students who did not use Hulu.

◯ The proportion of students who did use Hulu plus the proportion of students who did not use Hulu.

◯ The absolute value of the proportion of students who did use Hulu plus the proportion of students who did not use Hulu.

b) For the test statistic that you chose in part (a), what is the observed value of the statistic? Give your answer either as an exact decimal or a simplified fraction.

c) If the p-value of the hypothesis test is 0.053, what can we conclude, at the standard 0.05 significance level?

◯ We reject the null hypothesis.

◯ We fail to reject the null hypothesis.

◯ We accept the null hypothesis.

Before submitting your quiz, make sure your PID is on the front page and on the top right corner of page 3.